Differential Equations and Symmetries

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Abstract

This text serves as a self contained introduction to the use of symmetries in differential equations. First, we develop the tools necessary by a concise overview of Lie groups and algebras. Then, we use them to study the solvability of differential equations by integration. In the process, we'll find a general algorithm to find symmetries and how to use them to explicitly solve ODEs (when possible). We then draw links to Galois theory and solvability of polynomials. This text ends in an illustration of how to extend these methods to PDEs. We mainly follow the treatments given in [Hyd00] and [Olv93]. In general, this text focuses on the more intuitive and computational processes, as rigorous and theoretical approaches already exist. We assume that the reader is familiar with basic ODEs, vector calculus, linear algebra and group theory.

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1 Introduction and Motivation: First Order ODEs

(This section follows the first two chapters of [Hyd00])

What are symmetries of differential equations? Why do we care about them? This section's goal is to address these issues using a simple motivating example: first order ODEs. A general form is

$$y'(x) = \omega(x, y) \tag{1}$$

Here, $\omega(x, y)$ can be any smooth function of its arguments, such that the ODE has a unique solution on some domain. For simplicity, we will not consider more complicated ODEs that cannot be written in this form. As the section progresses we will see that in general, our example is very hard, if not impossible, to solve using standard ODE techniques. To begin with, we'll consider an easier sub-case. If $\omega(x, y)$ depends only on x, then the equation becomes

$$y'(x) = \omega(x) \tag{2}$$

This is much easier to solve. The answer is to simply integrate:

$$y(x) = C + \int \omega(x)dx \tag{3}$$

Where $C \in \mathbb{R}$ is an arbitrary constant determined by the initial conditions of the differential equation. The trick to solve (1) will be to try to make it look like (2), by using a carefully chosen change of variables $\big(r(x, y), s(x, y)\big)$, where r, s are the new coordinates. Then, having solved the differential equation with r and s, the answer can be found by reverting back to the original coordinates (x, y). Because of their importance, the special coordinates r and s are called *canonical coordinates*.

To find r and s, we first need to understand a property of (2). Looking at (3), notice that because $\omega(x)$ does not depend on y, a *solution curve* (the curve in $\mathbb{R}^2$ that $y(x)$ makes) can be translated along the y axis to give another solution. In other words, after a *point transformation* (essentially a change of variables) $(\hat{x}, \hat{y}) \mapsto (x, y + \varepsilon)$, for any $\varepsilon \in \mathbb{R}$, the differential equation is still satisfied for $(\hat{x}, \hat{y})$. Explicitly:

$$\begin{aligned} \hat{y}' = (y + \varepsilon)' &= y' \\ &= \omega(x) \quad \text{From the original equation, (2)} \\ &= \omega(\hat{x}) \quad \text{Since for this transformation, } x = \hat{x} \end{aligned} \tag{4}$$

Thus, when $y' = \omega(x)$ holds, then $\hat{y}' = \omega(\hat{x})$. Put differently, changing to coordinates $\hat{x}, \hat{y}$ doesn't change the equation. This is an example of a *symmetry*, and in particular a *Lie symmetry*. We'll see that an equation that has a translation symmetry is always in the form of 2, and hence can always be solved directly by integration.

More generally, any point transformation may be a symmetry as long as the following conditions are satisfied.

Definition 1.1 (Symmetry of a first order differential equation)**.** A point transformation
$\Gamma : (x, y) \mapsto \big(\hat{x}(x, y), \hat{y}(x, y)\big)$ is a symmetry if the following hold:

1. Γ is a *diffeomorphism*, meaning it's smooth and has a smooth inverse.
2. Γ preserves the differential equation, in the sense that

$$\hat{y}'(\hat{x}) = \omega(\hat{x}, \hat{y}) \quad \text{when} \quad y'(x) = \omega(x, y) \tag{5}$$

Symmetries that can be continuously varied are of special interest. The previous example is an instance of this phenomenon: ε is a real number and can be continuously varied.

Definition 1.2 (Lie symmetry of a first order differential equation)**.** A symmetry of a differential equation $\Gamma_\varepsilon : (x, y) \mapsto \big(\hat{x}_\varepsilon(x, y), \hat{y}_\varepsilon(x, y)\big)$ that depends smoothly on a parameter ε is a Lie symmetry if the following are satisfied:

1. Γ_0 is the identity symmetry, so $(\hat{x}_0, \hat{y}_0) = (x, y)$.
2. There exists an open neighborhood $N_0 \subset \mathbb{R}$ around 0 such that for all $\varepsilon \in N_0$, Γ_ε is a symmetry.
3. Composing two symmetries Γ_δ and Γ_ε is equivalent to applying the symmetry $\Gamma_{\delta+\varepsilon}$, for $\delta, (\varepsilon$ close to 0).
4. $(\hat{x}_\varepsilon, \hat{y}_\varepsilon)$ can be represented as a Taylor series in ε around 0:

$$\begin{aligned} \hat{x}_\varepsilon &= x + \varepsilon\xi(x, y) + O(\varepsilon^2) \\ \hat{y}_\varepsilon &= y + \varepsilon\eta(x, y) + O(\varepsilon^2) \end{aligned} \tag{6}$$

First order ODEs can have one or more Lie symmetries. However, symmetries are not limited to translations. The main idea is the following: we change variables to make the differential equation of the form (2). Ultimately, this is equivalent to changing variables so that in our new coordinate system the (possibly complicated) symmetry becomes a translation. The resulting system has the attractive quality of being solvable by integration.

To be more precise, say that

$$\Gamma : (x, y) \mapsto \big(\hat{x}_\varepsilon(x, y), \hat{y}_\varepsilon(x, y)\big)$$

is a Lie symmetry of some first order ODE, and that $\big(r(x, y), s(x, y)\big)$ is a change of coordinates. Define $(\hat{r}_\varepsilon, \hat{s}_\varepsilon) = \big(r(\hat{x}_\varepsilon, \hat{y}_\varepsilon), s(\hat{x}_\varepsilon, \hat{y}_\varepsilon)\big)$, the Lie symmetry in the new coordinates. We want $(\hat{r}_\varepsilon, \hat{s}_\varepsilon) = (r, s + \varepsilon)$, so translation in the new coordinates.

Next, to progress, it will be necessary to view symmetries from another perspective. Consider a point p in the plane that is being transformed under a Lie symmetry by a very small amount (so Γ_ε is applied for a very small ε). Since

Lie symmetries are smooth, then this infinitesimal motion will be approximately a small line going through p. Scaled up, this defines a vector at the point, and doing that at each point defines a vector field on the plane. This vector field associated to the transformation is called the *flow (or orbit)* of the point transformation.

This name comes from another interpretation of what the vector field is. Imagine that the parameter ε is time, and that at the point p there is a particle in a fluid whose motion is described by the transformation Γ_ε. Then the flow of Γ_ε at p would be the velocity vector of the particle going through it.

In both cases, the flow can be defined as

$$\left(\left.\frac{d\hat{x}_\varepsilon}{d\varepsilon}\right|_{\varepsilon=0}, \left.\frac{d\hat{y}_\varepsilon}{d\varepsilon}\right|_{\varepsilon=0}\right) \tag{7}$$

From equation (6), these are simply the first order coefficient of the Taylor expansion:

$$\begin{aligned}
&\left(\frac{d\hat{x}_\varepsilon}{d\varepsilon}, \frac{d\hat{y}_\varepsilon}{d\varepsilon}\right) \\
&= \left(\frac{d}{d\varepsilon}\Big[x + \varepsilon\xi(x,y) + O(\varepsilon^2)\Big], \frac{d}{d\varepsilon}\Big[y + \varepsilon\eta(x,y) + O(\varepsilon^2)\Big]\right) \\
&= \Big(\xi(x,y) + O(\varepsilon), \eta(x,y) + O(\varepsilon)\Big)
\end{aligned} \tag{8}$$

Evaluated at $\varepsilon = 0$, this gives

$$\left(\left.\frac{d\hat{x}_\varepsilon}{d\varepsilon}\right|_{\varepsilon=0}, \left.\frac{d\hat{y}_\varepsilon}{d\varepsilon}\right|_{\varepsilon=0}\right) = \big(\xi(x,y), \eta(x,y)\big) \tag{9}$$

We will omit the ε subscript from now on to lighten the notation. It will be implicit that transformations considered will depend on ε. The reason vector fields are convenient is because a translation along the y axis in vector field form is simply $(0,1)$, at all points of the plane. In other words, the coordinates we want should satisfy the following :

$$\left(\left.\frac{d\hat{r}}{d\varepsilon}\right|_{\varepsilon=0}, \left.\frac{d\hat{s}}{d\varepsilon}\right|_{\varepsilon=0}\right) = (0,1) \tag{10}$$

It's now possible to use the chain rule on the left hand side, using the fact that $(\hat{r},\hat{s}) = \big(r(\hat{x},\hat{y}), s(\hat{x},\hat{y})\big)$:

$$\begin{aligned}
&\left(\left.\frac{d\hat{r}}{d\varepsilon}\right|_{\varepsilon=0}, \left.\frac{d\hat{s}}{d\varepsilon}\right|_{\varepsilon=0}\right) \\
=&\left(\left.\frac{dr}{d\hat{x}}\frac{d\hat{x}}{d\varepsilon}\right|_{\varepsilon=0} + \left.\frac{dr}{d\hat{y}}\frac{d\hat{y}}{d\varepsilon}\right|_{\varepsilon=0}, \left.\frac{ds}{d\hat{x}}\frac{d\hat{x}}{d\varepsilon}\right|_{\varepsilon=0} + \left.\frac{ds}{d\hat{y}}\frac{d\hat{y}}{d\varepsilon}\right|_{\varepsilon=0}\right) \\
=&\left(\xi(x,y)\frac{dr}{d\hat{x}} + \eta(x,y)\frac{dr}{d\hat{y}}, \xi(x,y)\frac{ds}{d\hat{x}} + \eta(x,y)\frac{ds}{d\hat{y}}\right)
\end{aligned} \tag{11}$$

It should be reminded that because $\hat{r} = r(\hat{x}, \hat{y})$, then

$$\frac{dr(\hat{x}, \hat{y})}{d\hat{x}} = \frac{dr(x, y)}{dx} \tag{12}$$

And similarly for s. In 11, r has arguments $\hat{x}, \hat{y}$ (it was omitted for legibility), and the derivative is taken with respect to $\hat{x}, \hat{y}$ in the first and second term, respectively. Again, the same holds for s. That means equation 12 and the condition that the tangent vectors are $(0, 1)$ can be used to get two partial differential equations:

$$\begin{aligned} \xi(x, y)\frac{dr}{dx} + \eta(x, y)\frac{dr}{dy} &= 0 \\ \xi(x, y)\frac{ds}{dx} + \eta(x, y)\frac{ds}{dy} &= 1 \end{aligned} \tag{13}$$

Where now r and s are functions of x and y. This is an explicit equation linking a given symmetry of a differential equation, in the form of its vector field, and the canonical coordinates. These are "easily" solved for r, s using the method of characteristics (appendix A). The answer is then found by rewriting the original ODE in these coordinates, and then integrating. For now, it remains unclear how one should find the symmetries of some differential equation, a clearly crucial part of the process. This will be addressed in the next sections.

Example – Solving a first order ODE using symmetries

Consider the equation

$$y' = xy^2 - \frac{2y}{x} - \frac{1}{x^3} \tag{14}$$

The equation is invariant under Lie symmetry

$$(\hat{x}, \hat{y}) = (e^{\varepsilon}x, e^{-2\varepsilon}y) \tag{15}$$

The vector field is simple:

$$\xi(x, y) = x \qquad \eta(x, y) = -2y \tag{16}$$

So the equations to find the canonical coordinates are the following

$$\begin{aligned} x\frac{dr}{dx} - 2y\frac{dr}{dy} &= 0 \\ x\frac{ds}{dx} - 2y\frac{ds}{dy} &= 1 \end{aligned} \tag{17}$$

Using the method of characteristics (appendix A):

$$(r, s) = (x^2 y, \ln|x|) \tag{18}$$

This should make the original differential equation, 14, much easier to integrate. We can calculate the following:

$$\begin{aligned}\frac{ds}{dr} &= \frac{s_x + \omega(x,y)s_y}{r_x + \omega(x,y)r_y} \quad \text{(For a general ODE } y' = \omega(x,y)\text{)} \\ &= \frac{\frac{1}{x} + \left(xy^2 - \frac{2y}{x} - \frac{1}{x^3}\right)\cdot(0)}{2xy + \left(xy^2 - \frac{2y}{x} - \frac{1}{x^3}\right)\cdot(x^2)} \\ &= \frac{1}{r^2 - 1}\end{aligned} \tag{19}$$

Note that many algebra steps were skipped. Integrating gives

$$s = c - \frac{\ln(|r+1|) - \ln(|r-1|)}{2} \tag{20}$$

Where $c \in \mathbb{R}$ is an integration constant. Reverting back to the original coordinates gives (and with $(e^c)^2 \to c$)

$$y = \frac{c + x^2}{x^2(c - x^2)} \tag{21}$$

2 Lie Groups and Algebras

(This section draws from sections 1.2, 1.3 and 1.4 of [Olv93])

The above motivating situation can be generalized greatly, and that will be necessary to solve higher order ODEs and PDEs. Moreover, the formalism will be necessary to know when ODEs can be solved by integration in the first place, using something similar to Galois theory.

2.1 Symmetries as Smooth Groups

In the informal presentation, Lie symmetries were defined as smooth maps acting on the plane. Symmetries are naturally expressed in the language of groups, and smoothness in the language of differential geometry. Thus this motivates the following generalization of our previous definition of Lie symmetry:

Definition 2.1 (Lie Group)**.** An r-parameter Lie group is a group G which is also an r dimensional manifold, in such a way that for all $g, h \in G$, the group operation $m(g,h) = g \cdot h$

$$m : G \times G \to G \tag{22}$$

and the inversion operation $i(g) = g^{-1}$

$$i : G \to G \tag{23}$$

are smooth maps between manifolds.

For this text, r dimensional manifolds should just be viewed as things that can be parametrized smoothly with r parameters. A good picture to have in mind is a surface (for $r = 2$). For example, a sphere or a hyperboloid are 2 dimensional manifolds. Other examples include a curve in the plane (1 dimensional), $\mathbb{R}^n$ (n dimensional), or even the set of special orthogonal matrices, $SO(n)$, which forms a Lie group under matrix multiplication. What these have in common is that they can be described, or parametrized, using one or multiple open subsets of Euclidean space. These patches of Euclidean space that are used to describe the manifold are called *coordinates*, and usually we only work with them and not the manifold itself.

We are interested in Lie groups acting on $\mathbb{R}^n$. In that case, we'll consider functions that take in an element of the group, g, and a point in $\mathbf{x} \in \mathbb{R}^n$ and return another point in $\mathbb{R}^n$. We'll denote that function

$$\Psi_g(\mathbf{x}) = \left(\psi_g^1(\mathbf{x}), ..., \psi_g^n(\mathbf{x})\right) \tag{24}$$

Given the identity of the group, e, we'll also want $\Psi_e(\mathbf{x}) = \mathbf{x}$. Since we are going to use coordinates to represent points in the manifold, we will usually represent an element of the group g as coordinates $\boldsymbol{\varepsilon} = (\varepsilon^1, ..., \varepsilon^r)$ (assuming an r parameter group), and require that the identity e be represented by the zero vector $(0, ..., 0)$.

Examples – Lie Groups Acting on Euclidean Space

1. The rotation group in 2 dimensions, $SO(2)$, is a one parameter Lie group. It can be explicitly parametrized by

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{25}$$

   And its action on $\mathbb{R}^2$ is given by matrix multiplication of vectors. Note that $\theta = 0$ gives the identity matrix. It can also be shown that $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$, using trigonometric identities. Since addition of real numbers is commutative, $SO(2)$ is an abelian group.

2. In general, the group $SO(n)$ is characterized by two conditions. Let $R \in SO(n)$, then

$$R^T R = I \quad \det R = 1 \tag{26}$$

   Where I is the identity matrix. It's the natural generalization of rotations to n dimensional Euclidean space. The group itself has $n(n-1)/2$ parameters, and for this reason an explicit parametrization is quite complicated. In general, $SO(n)$ is not abelian.

3. As was mentioned before $\mathbb{R}^n$ is a manifold. Like all vector spaces it's also an abelian group with respect to addition. It has a natural action on $\mathbb{R}^n$ (so it acts on a copy of itself) by translations. Indeed, given a point in $\mathbf{x} \in \mathbb{R}^n$, then the group element $\mathbf{y} \in \mathbb{R}^n$ acts as $\mathbf{x} \mapsto \mathbf{x} + \mathbf{y}$.

4. A transformation that often appears when dealing with ODEs (like in the example of last section) is an arbitrary scaling of every dimension. In two dimensions, the group action is given by

$$\begin{pmatrix} e^{\lambda_1 \varepsilon} & 0 \\ 0 & e^{\lambda_2 \varepsilon} \end{pmatrix} \tag{27}$$

where this matrix acts on $\mathbb{R}^2$. The constants λ_1, λ_2 are arbitrary. The reason for using the exponential instead of just a scaling factor is because we want the identity matrix when $\varepsilon = 0$.

2.2 Tangent Spaces

In the motivating example, we used the infinitesimal behavior around $\varepsilon = 0$ to make things much easier. The equivalent for a more general Lie group is to determine how the group acts on points when the elements considered are infinitesimally close to the identity of the group. To formalize this concept, we'll need the *tangent space*.

The tangent space at a point p in an n-dimensional manifold M, is an n dimensional vector space over $\mathbb{R}$. Intuitively, it represents what the manifold looks like when zooming infinitely close around p. In the case of a surface, the tangent space corresponds to the tangent plane. For a curve, it's the one dimensional subspace spanned by the tangent vector (so a straight line).

More formally, consider a point p in an n dimensional manifold M. Let $(x^1, ..., x^n)$ be local coordinates of M around p, and let C be an arbitrary smooth curve going through p parametrized by $\gamma(t) = (x^1(t), ..., x^n(t))$, such that $\gamma(0) = p$. Then, define the tangent vector of this curve as simply the derivative:

$$\dot{\gamma}(t) = (\dot{x}^1(t), ..., \dot{x}^n(t)) \tag{28}$$

This is a vector in $\mathbb{R}^n$. Taking all possible curves passing through p and their derivatives, we get an n dimensional vector space of tangent vectors:

Definition 2.2 (Tangent space)**.** The tangent space at $p \in M$, denoted T_pM, is the vector space of tangent vectors of all curves passing through p.

The canonical basis of this vector space, $\mathbf{e}_1, ..., \mathbf{e}_n$, corresponds exactly to tangent vectors to the curves

$$(t, 0, ..., 0), \ ..., \ (0, ..., 0, t) \tag{29}$$

Thus, the basis depends on our coordinates, and in fact represents how our coordinates change around p. However, it can be shown that the vector space itself doesn't depend on what parametrization we use. Changing coordinates is equivalent to changing basis. Usually, the basis vectors are written $\partial/\partial x^1, ..., \partial/\partial x^n$, as partial differential operators. The reason for this notation is because there is a one to one correspondence between tangent vectors at p and directional

derivatives of scalar functions on M at p. So, in a sense, tangent vectors *are* directional derivative operators. Tangent vectors will be denoted

$$X = \sum_{i=1}^{n} a^i \frac{\partial}{\partial x^i} \tag{30}$$

Their action on scalar functions $f : M \to \mathbb{R}$ is as we'd expect:

$$X(f) = \sum_{i=1}^{n} a^i \frac{\partial f}{\partial x^i}, \tag{31}$$

the directional derivative of f in the direction of $(a^1, ..., a^r)$.

So, coming back to our original question, given an r parameter Lie group acting on $\mathbb{R}^n$, how does it behave when the elements considered are very close to the identity? First, we need explicit (local) coordinates for the group. Let these coordinates be $\boldsymbol{\varepsilon} = (\varepsilon^1, ..., \varepsilon^r)$. Let the action of the group be described by

$$\Psi_{(\varepsilon^1,...,\varepsilon^r)}(\mathbf{x}) = \left(\psi^1_{(\varepsilon^1,...,\varepsilon^r)}(\mathbf{x}), ..., \psi^n_{(\varepsilon^1,...,\varepsilon^r)}(\mathbf{x})\right) \tag{32}$$

Where ψ^i is a smooth function of its arguments, and represents the ith component of the group action. Now, fix $\mathbf{x}$ as an arbitrary point. Each coordinate is now a function from the Lie group to the real numbers. We can thus apply the vector basis $\partial/\partial\varepsilon^1, ..., \partial/\partial\varepsilon^r$ to see how Ψ changes along each parameter and evaluate at $\boldsymbol{\varepsilon} = 0$, to get the behavior around the identity. We get:

$$X_1 = \left.\frac{\partial \Psi_{(\varepsilon^1,...,\varepsilon^r)}(\mathbf{x})}{\partial \varepsilon^1}\right|_{\boldsymbol{\varepsilon}=0}, ..., X_r = \left.\frac{\partial \Psi_{(\varepsilon^1,...,\varepsilon^r)}(\mathbf{x})}{\partial \varepsilon^r}\right|_{\boldsymbol{\varepsilon}=0} \tag{33}$$

These vectors are now in $\mathbb{R}^n$, in fact they're an r dimensional subspace, at $\mathbf{x}$, where the basis is $\partial/\partial x^1, ..., \partial/\partial x^n$. They are called the *infinitesimal generators*, and usually, the vector space of generators of a group G is denoted $\mathfrak{g}$. In some sense, what we've done is push the action of $\partial/\partial\varepsilon^i$ through the group action to see how to acts on our target space. Another way to see the generators is as first order coefficients of a Taylor expansion of the group action at $\boldsymbol{\varepsilon} = 0$.

Example – The Generator of $SO(2)$

The matrix group $SO(2)$ represents rotations of 2D vectors. It's defined as matrices R that satisfy $R^T R = I$ and $\det(R) = 1$. An explicit parametrization of $SO(2)$ is given by

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{34}$$

Where θ is the parameter, playing the role of ε. It's a 1 parameter group, so we should expect it to have only one generator. Here, the group action

on the plane $(x, y) \in \mathbb{R}^2$ is

$$\Psi_\theta(x, y) = R(\theta)(x, y) \tag{35}$$

Applying the method described above, we get the derivative

$$\frac{\partial \Psi_\theta(x, y)}{\partial \theta} = (-x \sin\theta - y\cos\theta,\ x\cos\theta - y\sin\theta) \tag{36}$$

And evaluated at $\theta = 0$,

$$\left.\frac{\partial \Psi_\theta(x, y)}{\partial \theta}\right|_{\theta=0} = (-y, x) \tag{37}$$

Written in term of basis vectors $\partial/\partial x$ and $\partial/\partial y$, this is

$$X = -y\partial/\partial x + x\partial/\partial y \tag{38}$$

This is a one dimensional vector space at each point, the most general element is given by simply aX_1, where $a \in \mathbb{R}$ is an arbitrary constant. A single generator at every point is effectively a vector field, and can be easily visualized.

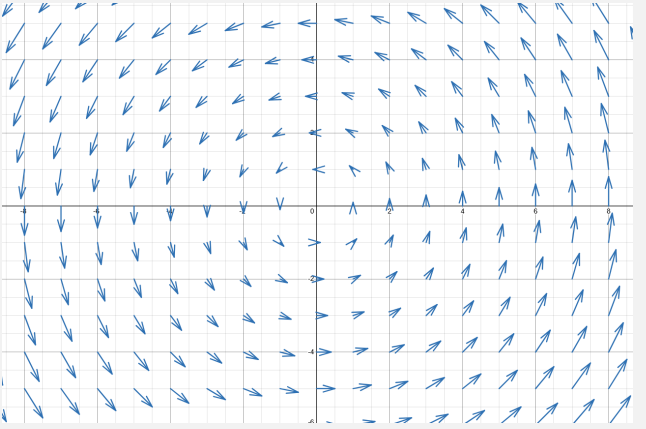


Figure 1: The generator of $SO(2)$ as a vector field.

This image highlights the following important idea: imagine taking a point and applying the group for all θ around 0. For definitiveness, take the point $(1, 0)$. Applying the group gives the curve

$$\theta \mapsto (x(\theta), y(\theta)) = (\cos\theta, \sin\theta) \tag{39}$$

This is called an *orbit* of $SO(2)$. In this case, it's part of the unit circle. At every point of that curve the tangent vector will be the infinitesimal generator. Indeed, the derivative is

$$(-\sin\theta, \cos\theta) = (-y(\theta), x(\theta)) \tag{40}$$

In general, for a one parameter group of transformation G, the orbits are the curves formed by applying G to a point and considering all other points that can be reached by symmetry. The infinitesimal generators are always tangent to these curves.
Given an initial point, it's also possible to recreate the orbit and the group action from only the generator, by trying to find curves that are tangent to it at all points. These curves are also called the *flow* of the vector field. Here we are seeing the first hints of a correspondence between the generators and the group, something we will consider in more detail later.

Example – The Generators of the Translation Group

Consider vectors in $\mathbb{R}^3$ and the group of transformations given by a general translation vector (a, b, c):

$$\Psi_{a,b,c}(x, y, z) = (x + a, y + b, z + c) \tag{41}$$

Applying the same method as before, we get 3 generators (as this is a group of 3 parameters)

$$X_1 = \partial/\partial x \qquad X_2 = \partial/\partial y \qquad X_3 = \partial/\partial z \tag{42}$$

The most general element of this 3 dimensional vector space is then given by

$$X = a^1 X_1 + a^2 X_2 + a^3 X_3 \tag{43}$$

Where the coefficients a^1, a^2, a^3 are independent elements of $\mathbb{R}$.

2.3 Lie Brackets and Algebras

Differential equations usually have more than one symmetry, and they can each be parametrized by multiple parameters. To consider all of these together, and study how different symmetries interact together, we'll need an additional structure on our vector space, making it an *algebra*.

Definition 2.3 (Lie algebra)**.** A Lie algebra $\mathfrak{g}$ is a vector space with a bilinear operation, the *Lie bracket* or *commutator*, $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \to \mathfrak{g}$. It's defined as the differential operator satisfying

$$[X_1, X_2](f) = \Big(X_1 X_2 - X_2 X_1\Big)(f) \tag{44}$$

By commutativity of differentiation, we can conclude that the second order term will be 0, making this another first order differential operator. The Lie bracket has the following properties

1. Bilinearity

$$\begin{aligned}[cX_1 + c'X_2, X_3] &= c[X_1, X_3] + c'[X_2, X_3] \\ [X_1, cX_2 + c'X_3] &= c[X_1, X_2] + c'[X_1, X_3]\end{aligned} \tag{45}$$

2. Skew-Symmetry

$$[X_1, X_2] = -[X_2, X_1] \tag{46}$$

3. Jacobi Identity

$$[X_1, [X_2, X_3]] + [X_3, [X_1, X_2]] + [X_2, [X_3, X_1]] = 0 \tag{47}$$

   In other words, adding all cyclic permutations of three vector fields will give zero.

We will require that the Lie bracket of two vectors in the Lie algebra is also in the Lie algebra (so closure under the Lie bracket). From this, we can deduce that all commutators of the basis vectors of an r dimensional algebra are of the following form

$$[X_i, X_j] = \sum_{k=1}^{r} c_{ij}^k X_k \tag{48}$$

Where the coefficients c_{ij}^k are called the *structure constants*. We will see that whether we can integrate a differential equation or not is entirely stored in the structure constants of the algebra of its symmetry group.

To define subalgebras, we need the following construction. Consider two subspaces (in the sense of a vector space) of the Lie algebra $\mathfrak{g}$, spanned by $M = \{X_1, ..., X_m\}$ and $N = \{Y_1, ..., Y_n\}$. Then let

$$[M, N] = \{[X_i, Y_j] : X_i \in M, Y_j \in N\} \tag{49}$$

Definition 2.4 (Subalgebra)**.** We want to define a subalgebra as a subspace that's closed under the Lie bracket. Let $M = \{X_1, ..., X_m\}$ be a basis of a subspace. M is a subalgebra if

$$[M, M] \subset M \tag{50}$$

Definition 2.5 (Ideal)**.** We can define another substructure. Let $L = \{X_1, ..., X_r\}$ be a basis of $\mathfrak{g}$. We say that M is an *ideal* if

$$[M, L] \subset M \tag{51}$$

Note that L itself is an ideal, as well as $\{0\}$. They are called *trivial ideals*. Ideals are automatically subalgebras, as $[M, M] \subset [M, L]$. The commutator of subspaces also gives us a very nice way to construct new algebras and ideals from old ones. Given a basis of the entire algebra L, define

$$L^{(1)} = [L, L] \tag{52}$$

We can show that $L^{(1)}$ is an ideal:

$$[L^{(1)}, L] = [[L, L], L] \subset [L, L] = L^{(1)} \tag{53}$$

The ideal $L^{(1)}$ may be equal to L, but in general it can be smaller. We can keep going and form another ideal

$$L^{(2)} = [L^{(1)}, L^{(1)}] \tag{54}$$

And so on, with

$$L^{(k+1)} = [L^{(k)}, L^{(k)}] \tag{55}$$

Where k is a positive integer. These ideals are called together a *derived series*. If eventually there comes an integer n such that $L^{(n)} = \{0\}$, then we say that the Lie algebra is *solvable*. As can be guessed, only ODEs having a solvable algebra from their symmetry group can be solved by integration. We will show that fact in later sections. This also explains why ideals are an important concept: they encode how symmetries are interlinked. Their definition essentially tells us if some generators can be "separated" from the entire algebra.

If we have an r dimensional solvable Lie algebra, then there exists a basis $\{X_1, ..., X_r\}$ such that the kth derived subalgebra is

$$L^{(k)} = \{X_1, ..., X_{r-k}\} \tag{56}$$

Such a basis is called a *canonical basis*. Equivalently, a basis is canonical if and only if the structure constants satisfy

$$c_{ij}^{k} = 0 \quad \forall i < j \leq k \tag{57}$$

2.4 A note on Galois Theory

Much of Lie theory, pioneered by Sophus Lie, is largely inspired by Galois theory. Though where Galois studied the solvability of polynomial equations, Lie theory sets out to study differential equations by analogous means. Despite the different settings, strikingly parallel notions prevail: solvability, chains of normal subgroups, and in the language of Lie theory ideals.

2.4.1 Field Extensions and the Galois Group

We start with the classical setup. Let $f \in \mathbb{Q}[x]$ be a polynomial of degree n with roots $\alpha_1, \ldots, \alpha_n \in \mathbb{C}$. The *splitting field* of f is the smallest field extension

$$\mathbb{Q} \subseteq K = \mathbb{Q}(\alpha_1, \ldots, \alpha_n) \subseteq \mathbb{C}.$$

Definition 2.6. The **Galois group** of f (or of the extension $K/\mathbb{Q}$) is

$$\mathrm{Gal}(K/\mathbb{Q}) := \{\, \sigma : K \xrightarrow{\sim} K \mid \sigma(q) = q \text{ for all } q \in \mathbb{Q} \,\},$$

the group of field automorphisms of K that fix $\mathbb{Q}$.

Every $\sigma \in \mathrm{Gal}(K/\mathbb{Q})$ permutes the roots $\{\alpha_1, \ldots, \alpha_n\}$, giving an injective homomorphism $\mathrm{Gal}(K/\mathbb{Q}) \hookrightarrow S_n$.

2.4.2 The Fundamental Theorem

The power of Galois theory rests on a perfect correspondence between the subgroup lattice of $\mathrm{Gal}(K/\mathbb{Q})$ and the lattice of intermediate fields.

Theorem 2.1 (Fundamental Theorem of Galois Theory)**.** Let $K/\mathbb{Q}$ be a finite Galois extension with group $G = \mathrm{Gal}(K/\mathbb{Q})$. There is an inclusion-reversing bijection

$$\{\text{subgroups } H \leq G\} \;\longleftrightarrow\; \{\text{intermediate fields } \mathbb{Q} \subseteq F \subseteq K\},$$

given by $H \mapsto K^H := \{x \in K \mid \sigma(x) = x,\ \forall\, \sigma \in H\}$. Further, $H \trianglelefteq G$ if and only if $K^H/\mathbb{Q}$ is itself a Galois extension.

2.4.3 Solvability by Radicals

The central question resolved by Galois was: *when can the roots of f be expressed using only field operations and radicals?*

Definition 2.7. A group G is **solvable** if there exists a subnormal series

$$\{e\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G$$

such that each quotient G_{i+1}/G_i is abelian.

Theorem 2.2 (Galois)**.** A polynomial $f \in \mathbb{Q}[x]$ is solvable by radicals if and only if its Galois group $\mathrm{Gal}(f)$ is a solvable group.

Example

The general quadratic $ax^2 + bx + c$ has Galois group $\mathbb{Z}/2\mathbb{Z}$, which is abelian, hence solvable, consistent with the familiar quadratic formula. The general quintic, however, has Galois group S_5. Since S_5 contains A_5 as a simple non-abelian normal subgroup, S_5 is *not* solvable, proving the Abel–Ruffini theorem: no general formula by radicals exists for degree ≥ 5.

2.4.4 The Bridge to Lie Theory

Lie's insight was to transplant this philosophy from polynomials to differential equations. In place of permutations of roots Lie considers *continuous symmetries*—transformations forming a smooth group, now called a **Lie group** G. The infinitesimal data of G is encoded in its **Lie algebra** $\mathfrak{g}$, and the role of normal subgroups is played by *ideals*.

Definition 2.8. A subspace $\mathfrak{h} \subseteq \mathfrak{g}$ is an **ideal** if $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$, i.e. $[X, Y] \in \mathfrak{h}$ for all $X \in \mathfrak{g}$, $Y \in \mathfrak{h}$.

A Lie algebra $\mathfrak{g}$ is called **solvable** if its *derived series*

$$\mathfrak{g}^{(0)} = \mathfrak{g}, \qquad \mathfrak{g}^{(k+1)} = [\mathfrak{g}^{(k)}, \mathfrak{g}^{(k)}]$$

eventually reaches zero: $\mathfrak{g}^{(N)} = 0$ for some N. This mirrors exactly the condition on the Galois group, and Lie's theorem (in the complex case) states that a Lie algebra is solvable if and only if its adjoint representation can be simultaneously upper-triangularised—a clean analogue of solvability by radicals.

Thus, the chain

$$\text{polynomial equations} \xrightarrow{\text{Galois}} \text{discrete groups} \xrightarrow{\text{Lie}} \text{continuous groups}$$

reveals how deeply the two theories echo one another, with solvability and chains of (normal) substructures as the common thread.

2.5 Relating the Group and the Algebra

As was hinted at previously, it's a fact that there's a one to one correspondence between a Lie group and its Lie algebra. We mentioned previously that the action of a one parameter group will trace a curve, and that at each point p this curve has tangent vector equal to the generator at p. We can use this fact to recover the group action.

If we have a one parameter group action $\Psi_\varepsilon(\mathbf{x}) : \mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$, and generator $X = \sum_{i=1}^n a^i(\mathbf{x})\partial/\partial x^i$, then:

$$\Big(a^1(\Psi_\varepsilon(\mathbf{x})), ..., a^n(\Psi_\varepsilon(\mathbf{x}))\Big) = \frac{\partial \Psi_\varepsilon(\mathbf{x})}{\partial \varepsilon} \tag{58}$$

This equation is exactly the statement that the tangent vector at a point of an orbit is the generator. To get back the group action we simply need to solve these n coupled differential equations with initial conditions given by

$$\mathbf{x} = \Psi_0(\mathbf{x}) \tag{59}$$

The process of solving that differential equation is often referred to as *exponentiation*, denoted:

$$\exp(\varepsilon X)\mathbf{x} = \Psi_\varepsilon(\mathbf{x}) \tag{60}$$

By considering the exponential as a power series, the above is not just notation but another valid way to solve the differential equation. Higher powers of X are considered higher order differential operators. For groups with r parameters, and thus having an r dimensional Lie algebra, we can recover any element of the group using multiple times the exponential map with some basis vectors:

$$\exp\Big(\varepsilon^1 X_1\Big) ... \exp\Big(\varepsilon^r X_r\Big)\mathbf{x} = \Psi_{(\varepsilon_1,...,\varepsilon_r)}(\mathbf{x}) \tag{61}$$

Using this, it's easy to see how any element of the algebra can be mapped to a group action, although the proof of this fact requires more work.

Example – Computing the 2D Rotation Group from its Generator

Consider the generator

$$X = -y\partial/\partial x + x\partial/\partial y \tag{62}$$

Let the group action on an arbitrary point (x, y) be $\Psi_\theta(x, y) = \big(x_\theta(x, y), y_\theta(x, y)\big)$. Then exponentiation is equivalent to solving the following equations

$$-y_\theta = x'_\theta \quad x_\theta = y'_\theta \tag{63}$$

Note that all derivatives are with respect to θ. The arguments (x, y) are also omitted for legibility. This is a linear system of ODEs with constant coefficients, so a well know method exists. The first step is to rewrite the equations as a single matrix equation:

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_\theta \\ y_\theta \end{pmatrix} = \begin{pmatrix} x'_\theta \\ y'_\theta \end{pmatrix} \tag{64}$$

Then, the solutions are of the form

$$\begin{pmatrix} x_\theta \\ y_\theta \end{pmatrix} = e^{\lambda\theta} \begin{pmatrix} u_1 \\ u_2 \end{pmatrix} \tag{65}$$

Where λ is an eigenvalue of the above matrix, and (u_1, u_2) is the eigenvector associated to of the matrix. The eigenvalues can be found using the characteristic polynomial, defined by

$$\det \begin{pmatrix} -\lambda & -1 \\ 1 & -\lambda \end{pmatrix} = 0 \tag{66}$$

This is easily solved: $\lambda = \pm i$, where $i^2 = 1$ is the imaginary unit. The eigenvectors can be computed using standard methods and are

$$\begin{pmatrix} i \\ 1 \end{pmatrix} \quad \begin{pmatrix} -i \\ 1 \end{pmatrix} \tag{67}$$

So the most general solution is given by

$$x_\theta = ic_1 e^{i\theta} - ic_2 e^{-i\theta} \quad y_\theta = c_1 e^{i\theta} + c_2 e^{-i\theta} \tag{68}$$

We now have to plug in the initial condition: at $\theta = 0$, $x_0 = x$ and $y_0 = y$, so

$$x = ic_1 - ic_2 \quad y = c_1 + c_2 \tag{69}$$

This system of equations gives

$$c_1 = \frac{x + iy}{2i} \qquad c_2 = -\frac{x - iy}{2i} \tag{70}$$

And so the solution, taking into account the initial conditions, is

$$x_\theta = \frac{x+iy}{2}e^{i\theta} + \frac{x-iy}{2}e^{-i\theta} \quad y_\theta = \frac{x+iy}{2i}e^{i\theta} - \frac{x-iy}{2i}e^{-i\theta} \tag{71}$$

Rearranging:

$$\begin{aligned} x_\theta &= x\Big(\frac{e^{i\theta}+e^{-i\theta}}{2}\Big) - y\Big(\frac{e^{i\theta}-e^{-i\theta}}{2i}\Big) \\ y_\theta &= x\Big(\frac{e^{i\theta}-e^{-i\theta}}{2i}\Big) + y\Big(\frac{e^{i\theta}+e^{-i\theta}}{2}\Big) \end{aligned} \tag{72}$$

And using Euler's formula for sine and cosine, we finally get

$$x_\theta = x\cos\theta - y\sin\theta \quad y_\theta = x\sin\theta + y\cos\theta \tag{73}$$

This is indeed the action of the rotation group on the plane! We can also use the exponential map directly. We then have to apply successive powers of the generator to (x, y). zeroth order:

$$1x = x \qquad 1y = y \tag{74}$$

First order:

$$\begin{aligned} \Big(-y\partial/\partial x + x\partial/\partial y\Big)(x) &= -y \\ \Big(-y\partial/\partial x + x\partial/\partial y\Big)(y) &= x \end{aligned} \tag{75}$$

Then, for the second order we take the result of what we got at first order and apply the generator again.

$$\begin{aligned} \Big(-y\partial/\partial x + x\partial/\partial y\Big)(-y) &= -x \\ \Big(-y\partial/\partial x + x\partial/\partial y\Big)(x) &= -y \end{aligned} \tag{76}$$

Third order:

$$\begin{aligned} \Big(-y\partial/\partial x + x\partial/\partial y\Big)(-x) &= y \\ \Big(-y\partial/\partial x + x\partial/\partial y\Big)(-y) &= -x \end{aligned} \tag{77}$$

By now the pattern is clearer. It can be shown that for k an integer,

$$X^{2k}x = (-1)^k x \qquad X^{2k+1}x = (-1)^{k+1}y \tag{78}$$

And similarly for y:

$$X^{2k}y = (-1)^k y \qquad X^{2k+1}y = (-1)^k x \tag{79}$$

With this, we can actually compute the power series, by splitting it into two. For x:

$$\begin{aligned}
&\exp(\theta X)x = \sum_{i=0}^{\infty} \frac{\theta^i}{i!} X^i(x) \\
&= \sum_{i=0}^{\infty} \frac{\theta^{2i}}{(2i)!} X^{2i}(x) + \sum_{i=0}^{\infty} \frac{\theta^{2i+1}}{(2i+1)!} X^{2i+1}(x) \\
&= \sum_{i=0}^{\infty} \frac{\theta^{2i}}{(2i)!} (-1)^i x + \sum_{i=0}^{\infty} \frac{\theta^{2i+1}}{(2i+1)!} (-1)^{i+1} y \\
&= x \sum_{i=0}^{\infty} \frac{\theta^{2i}}{(2i)!} (-1)^i - y \sum_{i=0}^{\infty} \frac{\theta^{2i+1}}{(2i+1)!} (-1)^i \\
&= x \cos\theta - y \sin\theta
\end{aligned} \tag{80}$$

The same can be done for y, and we get the exact same answer as before.

3 Symmetries of Differential Equations

(This section follows chapter 3 of [Hyd00])

We now have the necessary tools to analyze and find symmetries of any differential equation. Consider first a general ODE of the form

$$y^{(n)} = \omega(x, y, y', ..., y^{(n-1)}) \tag{81}$$

And a transformation of the plane given by

$$\Gamma : (x, y) \mapsto (\hat{x}, \hat{y}) \tag{82}$$

Transforming x and y will inevitably have an impact on all derivatives of y. For this reason, we will work in an extended space adapted to studying differential equations.

Definition 3.1 (Jet Space)**.** The Jet space is an $n+2$ dimensional space where each derivative is treated as an independent coordinate: $(x, y, y', ..., y^{(n)})$.

And we will consider the transformation as acting on this jet space.

$$\Gamma : (x, y, y', ..., y^{(n)}) \mapsto (\hat{x}, \hat{y}, \hat{y}', ..., \hat{y}^{(n)}) \tag{83}$$

This is called the nth *prolongation of* Γ, and here, $\hat{y}^{(k)}$ is defined as follows:

$$\hat{y}^{(k)} = \frac{d^k \hat{y}}{d\hat{x}^k} \tag{84}$$

Here the prime doesn't imply differentiation with respect to x, but with respect to $\hat{x}$. To compute the prolongation in terms of the jet space coordinates in practice, we can simply use the chain rule recursively:

$$\hat{y}^{(k)} = \frac{d\hat{y}^{(k-1)}}{d\hat{x}} = \frac{D_x\hat{y}^{(k-1)}}{D_x\hat{x}} \tag{85}$$

(From now on, we'll use the notation $D_x = \partial/\partial x + y'\partial/\partial y + y''\partial/\partial y' + ...$ when using the chain rule, for legibility and brevity. This operator is called the total derivative.) With all this, we can define what a symmetry is.

Definition 3.2 (Symmetry of an ODE)**.** Consider a transformation $\Gamma : (x, y) \mapsto (\hat{x}, \hat{y})$, and its nth prolongation $\Gamma : (x, y, y', ..., y^{(n)}) \mapsto (\hat{x}, \hat{y}, \hat{y}', ..., \hat{y}^{(n)})$. Γ is a symmetry if

$$\begin{aligned} \hat{y}^{(n)} &= \omega(\hat{x}, \hat{y}, \hat{y}', ..., \hat{y}^{(n-1)}) \\ \text{when } \ y^{(n)} &= \omega(x, y, y', ..., y^{(n-1)}) \ \text{ holds} \end{aligned} \tag{86}$$

This condition is intuitive but not easy to work with, because it's usually non-linear. For this reason, we will only work with Lie symmetries and use their infinitesimal generators.

Definition 3.3 (Lie Symmetry)**.** A Lie symmetry of an ODE is a Lie group acting on $\mathbb{R}^2$ preserving the differential equation. More precisely, an r-parameter Lie group

$$(x, y) \mapsto \Big(\hat{x}(x, y; \varepsilon_1, ..., \varepsilon_r), \hat{y}(x, y; \varepsilon_1, ..., \varepsilon_r)\Big) \tag{87}$$

is a symmetry if its nth prolongation is a symmetry for all $(\varepsilon_1, ..., \varepsilon_r)$ in a neighborhood of $\boldsymbol{\varepsilon} = 0$.

To go further and use the generators, we'll need to also extend them to the jet space. (We'll consider only a one parameter group for now) To first order in ε:

$$\begin{aligned} \hat{x} &= x + \varepsilon\xi + O(\epsilon^2), \\ \hat{y} &= y + \varepsilon\eta + O(\epsilon^2), \\ \hat{y}^{(k)} &= y^{(k)} + \varepsilon\eta^{(k)} + O(\epsilon^2), \quad k \geq 1 \end{aligned} \tag{88}$$

Where ξ, η, and in general $\eta^{(k)}$ are functions of x, y and derivatives. Note that the subscript (k) in $\eta^{(k)}$ is not for a derivative, it's simply to index the functions. To compute these, it's a recursive process. First,

$$\xi = \frac{d\hat{x}}{d\epsilon} \qquad \eta = \frac{d\hat{y}}{d\epsilon}. \tag{89}$$

The derivatives of the prolonged group action are computed as follows using the chain rule:

$$\hat{y}^{(k)} = \frac{d\hat{y}^{(k-1)}}{d\hat{x}} = \frac{D_x\hat{y}^{(k-1)}}{D_x\hat{x}} = \frac{y^{(k)} + \varepsilon D_x\eta^{(k-1)} + O(\varepsilon^2)}{1 + \varepsilon D_x\xi + O(\varepsilon^2)}. \tag{90}$$

Then plugging in $\hat{y}^{(k)} = y^{(k)} + \varepsilon\eta^{(k)} + O(\epsilon^2)$ and isolating for $\eta^{(k)}$ by considering only first order in ε,

$$\eta^{(k)} = D_x\eta^{(k-1)} - y^{(k)}D_x\xi. \tag{91}$$

Using these, it's possible to extend the generator of a given symmetry $X = \xi\partial/\partial x + \eta\partial/\partial y$ to the jet space

$$\mathrm{pr}^{(n)}X = \xi\partial/\partial x + \eta\partial/\partial y + ... + \eta^{(n)}\partial/\partial y^{(n)} \tag{92}$$

If we were to follow points in the jet space after transformation, the above expression will tell us where each point is going. With this, we can formulate the symmetry condition in the prolonged tangent space at the identity. This is done by taking the derivative with respect to ε on both sides of the first symmetry condition, equation 86, and looking at first order in ε.

$$\begin{aligned} &\mathrm{pr}^{(n)}X\left(\hat{y}^{(n)} - \omega(\hat{x}, \hat{y}, ..., \hat{y}^{(n-1)})\right) = 0 \\ \iff\ &\eta^{(n)} - \left(\xi\omega_x + \eta\omega_y + ... + \eta^{(n-1)}\omega_{y^{(n-1)}}\right) = 0 \end{aligned} \tag{93}$$

When $y^{(n)} = \omega(x, y, ..., y^{(n-1)})$ holds. This linearized symmetry condition is, in general, an extremely long expression. However, it's very useful as we can use it to find symmetries of an ODE by letting ξ, η be arbitrary and solving for them. Note that multi-parameter symmetries can also be found using this condition, as nothing really changes in the derivation if there are more dimensions.

The fact that there's a correspondence between a Lie group and its algebra is of great importance here. Indeed, in this derivation we used an expansion to first order in ε around 0. There was no guarantee that higher order terms could be thrown away. But using the fact that the entire group action can be recreated from the generators makes the linearized symmetry condition true.

Example – Finding Symmetries of a Second Order ODE

Consider the ODE

$$y'' = \frac{y'^2}{y} - y^2. \tag{94}$$

First, we need to compute $\mathrm{pr}^{(2)}X$. We'll leave ξ and η general, since they're what we're looking for. The first prolongation is

$$\eta^{(1)} = \eta_x + y'\eta_y - y'(\xi_x + y'\xi_y) = \eta_x + (\eta_y - \xi_x)y' + (\xi_y)y'^2. \tag{95}$$

And the second prolongation is

$$\eta^{(2)} = \eta_{xx} + (2\eta_{xy} - \xi_{xx})y' + (\eta_{yy} - 2\xi_{xy})y'^2 - (\xi_{yy})y'^3 + (\eta_y - 2\xi_x - \xi_y y')y''. \tag{96}$$

We also need to compute derivatives of $\omega(x, y, y') =:$

$$\begin{aligned}\omega_x &= 0,\\ \omega_y &= -\frac{y'^2}{y^2} - 2y,\\ \omega_{y'} &= \frac{2y'}{y}.\end{aligned} \tag{97}$$

Finally, putting everything together in the symmetry condition gives

$$\begin{aligned}&\eta^{(2)} = \Big(\frac{-y'^2}{y^2} - 2y\Big)\eta + \Big(\frac{2y'}{y}\Big)\eta^{(1)}\\ \iff &\eta_{xx} + (2\eta_{xy} - \xi_{xx})y' +\\ &(\eta_{yy} - 2\xi_{xy})y'^2 - (\xi_{yy})y'^3 + (\eta_y - 2\xi_x - \xi_y y')y''\\ &= \Big(\frac{-y'^2}{y^2} - 2y\Big)\eta + \Big(\frac{2y'}{y}\Big)\eta_x + y'\eta_y - y'(\xi_x + y'\xi_y)\\ &= \eta_x + (\eta_y - \xi_x)y' + (\xi_y)y'^2\end{aligned} \tag{98}$$

After some algebra, and matching powers of y', we arrive at the following partial differential equations:

$$\begin{aligned}\xi_{yy} + \frac{1}{y}\xi_y &= 0\\ \eta_{yy} - 2\xi_{xy} - \frac{1}{y}\eta_y + \frac{1}{y^2}\eta &= 0\\ 2\eta_{xy} - \xi_{xx} + 3y^2\xi_y - \frac{2}{y}\eta_x &= 0\\ \eta_{xx} - y^2(\eta_y - 2\xi_x) + 2y\eta &= 0\end{aligned} \tag{99}$$

Initially this can look much harder than the original ODE, but this is actually quite easy to solve (although that's not always the case). The first equation gives something of the form

$$\xi = A(x)\ln|y| + B(x) \tag{100}$$

And the second equation gives

$$\eta = A'(x)y\big(\ln|y|\big)^2 + C(x)y\ln|y| + D(x)y \tag{101}$$

Where A, B, C, D are unknown functions to be determined with the other equations. The third equation gives

$$3A''(x)\ln|y| + 3A(x)y + 2C'(x) - B''(x) = 0 \tag{102}$$

From this it must follow that

$$A(x) = 0 \qquad B''(x) = 2C'(x) \tag{103}$$

And finally the last equation gives

$$C(x)y^2 \ln|y| + C''(x)y \ln|y| + (2B'(x) - C(x) + D(x))y^2 + D''(x)y = 0 \tag{104}$$

Which can be separated into the following conditions

$$C(x) = 0 \qquad D(x) = -2B'(x) \qquad D''(x) = 0 \tag{105}$$

This implies that $B(x) = c_1 + c_2 x$ and $D(x) = -c_2$. Finally, we have the most general symmetry:

$$\xi = c_1 + c_2 x \qquad \eta = -2c_2 y \tag{106}$$

There are two degrees of freedom here, so the tangent space is two dimensional. The most general generator is of the form $X = c_1 X_1 + c_2 X_2$, and from this we can compute the two basis vectors:

$$X_1 = \partial/\partial x \qquad X_2 = x\partial/\partial x - 2y\partial/\partial y \tag{107}$$

It can be interesting to use the exponential map, to see what the transformations looks like. For the first generator, it's equivalent to solving the following coupled ODEs.

$$\frac{d\hat{x}}{d\varepsilon} = 1 \qquad \frac{d\hat{y}}{d\varepsilon} = 0 \tag{108}$$

With initial conditions $(\hat{x}(0), \hat{y}(0)) = (x, y)$. This is clearly the transformation

$$\big(\hat{x}_\epsilon(x,y), \hat{y}_\epsilon(x,y)\big) = \big(x + \epsilon, y\big) \tag{109}$$

This makes sense: our original equation didn't depend on x. The second basis vector gives the following equations

$$\frac{d\hat{x}}{d\varepsilon} = x \qquad \frac{d\hat{y}}{d\varepsilon} = -2y \tag{110}$$

With, again, initial conditions $(\hat{x}(0), \hat{y}(0)) = (x, y)$. This is also easy to solve and gives

$$\big(\hat{x}_\epsilon(x,y), \hat{y}_\epsilon(x,y)\big) = \big(xe^{\epsilon}, ye^{-2\epsilon}\big) \tag{111}$$

4 Differential Invariants

(This section comes from the subsection "Differential Invariants" of section 2.5, from [Olv93])

So far, we've only looked at how to find symmetries for a given ODE. However, we can do the opposite procedure, which is to find the most general ODE that satisfies a given symmetry. For this, we need something called a *differential*

invariant: an expression including x, y and derivatives of y that stays constant under the group action. The concept of a differential invariant will also be useful when trying to solve higher order ODEs. In this section, we will not show proofs and stay away from too many details. We encourage the reader to deepen their understanding by reading chapter 2 of [Olv93].

Definition 4.1 (Differential Invariant)**.** Let G be a group of transformations acting on the plane (x, y). An nth order differential invariant of G is a smooth function f from the jet space to $\mathbb{R}$, such that it's invariant under the prolonged group action of G:

$$f\Big(\text{pr}^{(n)}g \cdot (x, y, y', ..., y^{(n)})\Big) = f(x, y, y', ..., y^{(n)}) \tag{112}$$

Let $F(x, y, y', ..., y^{(n)})$ and $H(x, y, y', ..., y^{(n)})$ be two differential invariants of nth order of a group of transformation G. Then there's a way to create higher order differential invariants. Indeed,

$$\frac{dH}{dF} = \frac{D_x H}{D_x F} \tag{113}$$

is a differential invariant of order $n + 1$.

Before going any further, we need to make a detour and define functional dependence. For a given group G, we might be interested to know if, in a sense, similar to linear algebra, there's a "basis" of functions $f_1, ..., f_n$, themselves differential invariants, such that every other differential invariant is of the form $F(f_1, ..., f_n)$. In other words, can we characterize every possible differential invariants using only a small set of functions? In general, the answer is yes. We say that these functions are a *complete set* and *functionally independent*. The precise definition of functional dependence is not that important here, the important part to understand is that we can build all differential invariants using only our "basis".

We can say something about the maximum number of functionally independent differential invariants of a group of transformation G with r parameters acting on $\mathbb{R}^n$. In general, there are $n - r$ invariants. For instance, if we have a one parameter group acting on the plane, then there's only one functionally independent differential invariant. If we take the first prolongation, then there are two, and so on.

Consider now differential invariants of order 0 and 1: $u(x, y)$ and $w(x, y, y')$. Further assume that they are a complete set of functionally independent invariants of the first prolongation of G. Then, the derivatives

$$u, w, \frac{dw}{du}, ..., \frac{d^{n-1}w}{du^{n-1}} \tag{114}$$

are a complete set of functionally independent invariants for the nth prolongation of G. For this reason, the differential invariants of order 0 and 1 are called *fundamental differential invariants*, because it's possible to find all others using them. The most general differential equation of order n that has G as a

symmetry is then an arbitrary smooth function of all differential invariants of order $n-1$ and below.

This leads to the question of how to find, in practice, the fundamental differential invariants of a group G. As usual, it's way easier to work in the tangent space at the identity, so consider the generator (with first prolongation only)

$$\mathrm{pr}^{(1)} X = \xi\, \partial/\partial x + \eta\, \partial/\partial y + \eta^{(1)}\, \partial/\partial y' \tag{115}$$

Let $f(x, y, y')$ be the differential invariant that we're searching for. Then, it needs to satisfy

$$\mathrm{pr}^{(1)} X(f) = 0 \implies \xi\, f_x + \eta\, f_y + \eta^{(1)}\, f_{y'} = 0 \tag{116}$$

Where subscripts indicate derivation. This condition comes from the fact that we want our differential invariant to stay constant along the flow of the group action. Since the generator at a point represents the direction of the flow, then the directional derivative along the generator should always be 0. That's the expression above. In general, there are two linearly independent solutions, as we'd expect. These solutions can be found using the method of characteristics.

To find differential invariants of an entire subalgebra with a higher dimension, then we can apply the condition above but to all basis generators and get multiple coupled equations.

Example – The rotation group $SO(2)$

The rotation group of the plane, $SO(2)$, has generator

$$X = -y\, \partial/\partial x + x\, \partial/\partial y \tag{117}$$

Which admits the following first prolongation

$$\mathrm{pr}^{(1)} X = -y\, \partial/\partial x + x\, \partial/\partial y + \big(1 + (y')^2\big)\partial/\partial y' \tag{118}$$

So, the equation to solve to find invariants $f(x, y, y')$ is the following

$$0 = -y\, f_x + x\, f_y + \big(1 + (y')^2\big) f_{y'} \tag{119}$$

It's important to remember, we're work in the jet space and derivatives of y are simply coordinates. The two solution are

$$f_1(x, y) = x^2 + y^2 \qquad f_2(x, y, y') = \frac{x + y' y}{\sqrt{1 + (y')^2}} \tag{120}$$

Thus, the most general first order differential equation that has a rotation symmetry is a function of both of these expressions. Notice that the first expression is just a circle, as one would expect.

5 Solving Differential Equations Using Symmetries

(This section comes from section 2.5 of [Olv93])

We're finally ready to solve ODEs. The result that allows us to integrate differential equation is that if an ODE admits a one parameter symmetry, then it's possible to reduce the order of the differential equation by one using integration. The process is similar to what we did in the first section. To see how this works, consider a general ODE

$$\tilde{\Omega}(x, y, y', ..., y^{(n-1)}, y^{(n)}) = 0 \tag{121}$$

Consider Ω as simply $y^{(n)} - \omega(x, y, ...)$. It'll make everything easier in this section to notate things that way. Suppose this equation has symmetry generator

$$X = \xi \frac{\partial}{\partial x} + \eta \frac{\partial}{\partial y} \tag{122}$$

We then change into canonical coordinates (r, s). As a reminder, canonical coordinates are chosen so that the symmetry generator becomes this simple expression:

$$X = \frac{\partial}{\partial s} \tag{123}$$

To find these coordinates, it's quite simple. We want two things: $Xs = (\partial/\partial s)s = 1$ and $Xr = (\partial/\partial s)r = 0$. These two equations, with equation 122, become

$$\begin{aligned} \xi \frac{\partial}{\partial x} s + \eta \frac{\partial}{\partial y} s &= 1 \\ \xi \frac{\partial}{\partial x} r + \eta \frac{\partial}{\partial y} r &= 0 \end{aligned} \tag{124}$$

These can be easily solved (usually) using the method of characteristics. Once we have our new coordinates, we can re-express our ODE, equation 121, using the chain rule:

$$\Omega(r, s, s', ..., s^{(n)}) = 0 \tag{125}$$

where Ω is a function of r, s and derivatives of s with respect to r. X is still a symmetry, since it's the same equation but in different coordinates. Before applying it, however, we need to compute its prolongation, which takes a particularly simple form:

$$\mathrm{pr}^{(n)} X = \frac{\partial}{\partial s} \tag{126}$$

which implies, from the linearized symmetry condition, that

$$\frac{\partial}{\partial s} \Omega(r, s, s', ..., s^{(n)}) = 0 \tag{127}$$

So, in fact, the equation has reduced to

$$\Omega(r, s', ..., s^{(n)}) = 0 \tag{128}$$

And letting $z = s'$,

$$\Omega(r, z, z', ..., z^{(n-1)}) = 0 \tag{129}$$

Where primes indicate derivatives of z with respect to r. And so we have reduced the order by 1. If $z(r)$ is found, then the solution to the original equation (in canonical coordinates) is given by integrating

$$s = \int z(r)dr + C \tag{130}$$

We can apply this process recursively as long as we have one parameter symmetries, and then revert back to our original coordinates.

The above is useful if we have one parameter symmetries, but what if we have an r-parameter symmetry group instead? Intuitively, we'd think that we can reduce the order by r, and that's correct. However, we saw in the previous example that the integration step crucially depends on the fact that the symmetry has only one parameter. Even though we can reduce the order of the equation, it's possible that we cannot relate the solution of the new, reduced equation to the solution of our original equation. We might also think to use one parameter subgroups in succession and integrate r times. This actually doesn't work in general because using some subgroups will make it impossible to use others: they are somehow interlinked.

We deduce that the group of symmetries has to be special. Indeed, a differential equation of order n can have its order reduced by an r parameter group of symmetries using integration only if its algebra is solvable, as was mentioned before. This is because they allow us to isolate one parameter subgroups and use the method above. Let's see how this works.

Consider again the following general differential equation

$$\Omega(x, y, y', ..., y^{(n)}) = 0 \tag{131}$$

Let $\mathfrak{g}$ be the algebra of an r parameter symmetry group and further assume that it is solvable. We then have a canonical basis of generators $\{X_1, ..., X_r\}$ such that each subalgebra in the chain

$$\mathfrak{g}^{(r)} = \{0\} \subset \mathfrak{g}^{(r-1)} = \text{span}\{X_1\} \subset ... \subset \mathfrak{g}^{(0)} = \mathfrak{g} = \text{span}\{X_1, ..., X_r\} \tag{132}$$

is an ideal of the subalgebra coming just before. First, we can use the subalgebra $\mathfrak{g}^{(r-1)}$ to reduce the order by one. The method was described previously, we can simply take canonical coordinates (r, s), let $z = s'$, and the equation becomes:

$$\Omega^{(1)}(r, z, z', ..., z^{(n-1)}) = 0 \tag{133}$$

Now to use $\mathfrak{g}^{(r-2)}$. We've already used X_1, and by our choice of coordinates the equation is now invariant under X_1 (i.e. r and z are fundamental differential invariants of X_1, since it has prolongation $\partial/\partial s$ and thus $X_1 r = X_1 z = 0$).

What we'd like to do next is write X_2 using r, z. But remember, X_2 is a generator defined on the entire jet space $(x, y, y', ...)$, or equivalently, $(r, s, s', ...)$. There is no guarantee that we can write X_2 without s, using only r, z and its derivatives. So when is that possible? It turns out that it's exactly when $[X_1, X_2] \in \text{span}(X_1)$.

Indeed, assume that $[X_1, X_2] = kX_1$, for some $k \in \mathbb{R}$. That means that for any differential invariant f of X_1,

$$\begin{aligned} & X_1(X_2(f)) - X_2(\underbrace{X_1(f)}_{0}) = k\underbrace{X_1(f)}_{0} \\ \implies & X_1(X_2(f)) = 0 \end{aligned} \tag{134}$$

So $X_2(f)$ is also a differential invariant of X_1. But the entire differential equation is made up of differential invariants of X_1, since by the linearized symmetry condition $X_1\Omega^{(1)} = 0$. This fact is very important, because it follows that even by removing s, X_2 still has a well defined action on any equation written with the coordinates r, z. In other words, the action of X_2 on r, z stays within the coordinates r, z. It's then possible to find canonical coordinates of X_2 in those coordinates, and forget about s.

This process continues with other subalgebras until they've all been used in order. We can show inductively that the above process can be applied again and again:

Say that we have used the subalgebra $\mathfrak{g}^{(r-k)}$ to reduce the order by k, and have written the ODE in terms of differential invariants r_k, z_k of $\mathfrak{g}^{(r-k)}$:

$$\Omega^{(k)}(r_k, z_k, z_k', ..., z_k^{(n-k)}) = 0 \tag{135}$$

Consider now $\mathfrak{g}^{(r-k-1)}$, the next subalgebra in the derived series. We know that there exists some X_{k+1} such that $X_{k+1} \in \mathfrak{g}^{(r-k-1)}$, but $X_{k+1} \notin \mathfrak{g}^{(r-k)}$. We will use this generator to further reduce the order by one using integration.

Once again we wish to write X_{k+1} in terms of differential invariants of $\mathfrak{g}^{(r-k)}$. That's possible because $[\mathfrak{g}^{(r-k-1)}, \mathfrak{g}^{(r-k-1)}] \subset \mathfrak{g}^{(r-k)}$: just like before, this implies that if f is a differential invariant of $\mathfrak{g}^{(r-k)}$ then $X_{k+1}(f)$ is also a differential invariant of $\mathfrak{g}^{(r-k)}$, hence the action of X_{k+1} on r_k, z_k is well defined within that coordinate system and we can find canonical coordinates, which reduces the order.

In summary, solvable algebras are defined this way because they're the minimum structure necessary to successively find canonical coordinates and apply the method for one parameter groups.

Example – Solving an ODE with a Multi-parameter Symmetry Group

(This is example 2.62 from [Olv93])
The ODE we'll look at is

$$x^2 y_{xx} = H(xy_x - u) \tag{136}$$

It can be shown that this ODE admits the two parameter symmetry group

$$(x, y) \mapsto (\lambda x, y + \varepsilon x), \quad \varepsilon \in \mathbb{R}, \lambda \in \mathbb{R}_{\geq 0} \tag{137}$$

Then, just taking the derivatives with respect to ε and λ gives us the generators

$$X_1 = x\partial_y \quad X_2 = x\partial_x \tag{138}$$

Where X_1 comes from differentiation with respect to ε, and X_2 from differentiation with respect to λ. Let's now compute the derived series to make sure this is solvable by integration. Let $\phi(x, y)$ be a test function. Then we have

$$\begin{aligned} &[X_1, X_2](\phi(x, y)) \\ =&\Big(x\partial_y(x\phi_x) - x\partial_x(x\phi_y)\Big) \\ =&x^2\phi_{xy} - x\Big(\phi_y + x\phi_{yx}\Big) \\ =&-x\phi_y = -X_1\phi \end{aligned} \tag{139}$$

We can conclude that $[X_1, X_2] = -X_1$. Then, $[X_1, X_1] = 0$. This means that we indeed have a derived series that goes to the trivial ideal, and thus that our algebra is solvable. In details, we have

$$\mathfrak{g}^{(2)} = \{0\} \subset \mathfrak{g}^{(1)} = \text{span}\{X_1\} \subset \mathfrak{g}^{(0)} = \mathfrak{g} = \text{span}\{X_1, X_2\} \tag{140}$$

According to our procedure, first we need to find the canonical coordinates of X_2. This can be done as usual by imposing $X_1 r = 0,\ X_1 s = 1$, which is equivalent to the following two equations

$$xr_y = 0 \quad xs_y = 1 \tag{141}$$

These are easily solved to give $r = f(x)$ and $s = y/x + g(x)$. Since $f(x)$ and $g(x)$ are arbitrary, then we can choose them to be simple. Let $f(x) = x$ and $g(x) = 0$. Note that $f(x)$ cannot be 0 because because otherwise we would have only one functionally independent function. The next step is to compute ds/dr and d^2s/dr^2, and rewrite the ODE in those coordinates.

$$\frac{ds}{dr} = \frac{d}{dx}\Big(\frac{y}{x}\Big) = \frac{xy_x - y}{x^2} \tag{142}$$

This is a first order differential invariant of the first prolongation of X_1. Since any function of the differential invariants is also a differential invariant, then define $w = x^2 ds/dr = xy_x - y$ (this will be our second fundamental differential invariant with r). Then:

$$\frac{dw}{dr} = \frac{dw}{dx} = y_x + xy_{xx} - y_x = xy_{xx} \tag{143}$$

From now on we'll only use x and not r, as they're equal. Using these coordinates, the differential equation becomes

$$x\frac{dw}{dx} = H(w) \tag{144}$$

We have successfully reduced the order of our equation. This is now a separable equation, which can be solved for w as follows:

$$\int \frac{dw}{H(w)} = \ln|x| + C_1 \tag{145}$$

For $C_1 \in \mathbb{R}$. If we find w as a function of x, then we can find y by solving the equation

$$w(x) = xy_x - y \implies \frac{w(x)}{x} = y_x - \frac{1}{x}y \tag{146}$$

This can be done by the integrating factors method. Indeed, first we compute

$$\mu(x) = \exp\Big(-\int \frac{1}{x}\Big) = \exp(-\ln|x|) = \frac{1}{\exp(\ln|x|)} = \frac{1}{|x|} \tag{147}$$

Then the general solution is found by the following formula:

$$y = C_2 + \frac{1}{\mu(x)}\int \frac{1}{x}\mu(x)w(x)dx = C_2 + |x|\int \frac{w(x)}{x^2}dx \tag{148}$$

For $C_2 \in \mathbb{R}$.

6 PDEs

(This section comes from chapter 8 and 9 of [Hyd00])

For PDEs, everything that we did can be almost immediately transferred over. Let $f(x^1, ..., x^n) : \mathbb{R}^n \to \mathbb{R}$ be a scalar function, and denote its derivatives using $f_k = \partial f/\partial x^k$. A general PDE can be written as

$$\Delta(x^1, ..., x^n, f, f_1, f_2, ..., f_{11}, ...) = 0 \tag{149}$$

Where $\Delta(x^1, ..., x^n, f_1, f_2, ...)$ is a function of the coordinates, the function, and its derivatives. Denote a transformation of $\mathbb{R}^n$ and f by

$$\Gamma : (x^1, ..., x^n, f) \mapsto (\hat{x}^1, ..., \hat{x}^n, \hat{f}) \tag{150}$$

Once again, we need to define the jet space and the prolongation of this transformation. The jet space is now $(x^1, ..., x^n, f_1, ..., f_n, f_{11}, ...)$, where all original

coordinates and possible combinations of derivatives up to a certain degree are now coordinates. We can define the total derivative on the jet space as follows

$$\begin{aligned} D_1 &= \partial/\partial x^1 + f_1\partial/\partial f + f_{11}\partial/\partial f_1 + ... + f_{1n}\partial/\partial f_n + ... \\ &... \\ D_n &= \partial/\partial x^n + f_n\partial/\partial f + f_{n1}\partial/\partial f_n + ... + f_{nn}\partial/\partial f_n + ... \end{aligned} \tag{151}$$

The transformed jet space coordinates are again just defined recursively. For first order, we have $\hat{f} = \hat{f}(\hat{x}^1, ..., \hat{x}^n)$, and can use the chain rule:

$$D_i\hat{f} = \sum_{j=1}^{n} D_i\hat{x}^j f_j \tag{152}$$

This is a linear system of equations, so we can use Cramer's rule:

$$\hat{f}_i = \frac{1}{J}\det\begin{pmatrix} D_1\hat{x}^1 & \dots & D_1\hat{f} & \dots & D_1\hat{x}^n \\ \vdots & & \vdots & & \vdots \\ D_n\hat{x}^1 & \dots & \underbrace{D_n\hat{f}}_{i\text{th column}} & \dots & D_n\hat{x}^n \end{pmatrix} \tag{153}$$

Where J is defined as the determinant of the matrix whose components are $D_i\hat{x}^j$. It's essentially the Jacobian determinant for the transformed coordinates. This process can be applied recursively to find the next higher order prolongations. With this, the definition of a symmetry is as follows:

Definition 6.1 (Symmetry of a PDE)**.** The transformation Γ is a point symmetry if

$$\Delta(\hat{x}^1, ..., \hat{x}^n, \hat{f}, \hat{f}_1, \hat{f}_2, ..., \hat{f}_{11}, ...) = 0 \tag{154}$$

when $\Delta(x^1, ..., x^n, f, f_1, f_2, ..., f_{11}, ...) = 0$ holds.

As before, we can deduce from the above another symmetry condition for the tangent space at the identity.

Definition 6.2 (Linearized symmetry condition for PDEs)**.** Let X be the generator of a symmetry, and take its nth prolongation, $\text{pr}^n X$. Then X is a symmetry if

$$\text{pr}^n X(\Delta) = 0 \quad \text{when} \quad \Delta = 0 \tag{155}$$

In practice, the prolongation of X is a very long expression, just like for ODEs. To compute it, it's also a recursive process. First, let's put symbols to the generators:

$$\left.\frac{d\hat{x}^k}{d\varepsilon}\right|_{\varepsilon=0} = \xi^k(x^1, ..., x^n, f) \qquad \left.\frac{d\hat{f}}{d\varepsilon}\right|_{\varepsilon=0} = \eta(x^1, ..., x^n, f) \tag{156}$$

Next, to first order (consider this a definition of η^k):

$$\hat{f}_k = f_k + \varepsilon\eta^k + O(\varepsilon^2) \tag{157}$$

We can isolate η^k by differentiating with respect to ε, setting it to 0, and using 153:

$$\eta^k = D_k \eta - \sum_{i=1}^{n} f_i D_k \, \xi^i \tag{158}$$

More generally, consider $\eta^{k_1 \dots k_m}$. From it we can compute higher order prolongation coefficients:

$$\eta^{k_1 \dots k_m j} = D_j \eta^{k_1 \dots k_m} - \sum_{i=1}^{n} f_{k_1 \dots k_m i} D_j \, \xi^i \tag{159}$$

With these definitions alone we can go pretty far. Indeed, once we have found the symmetries as group actions and a single solution, we can reconstruct many other solutions, called *group-invariant solutions*. The basic idea is that if we transform a solution using a symmetry, then we should get back another solution. Let's see how this works.

Consider a general symmetry of the following form

$$(\hat{x}, \hat{f}) = g \cdot (x, f) = \Big(X_g(x, f), F_g(x, f) \Big) \tag{160}$$

Where X, F are smooth functions of their arguments. Also note that x and $\hat{x}$ could represent vectors. We have to be careful and make a distinction between $f(x)$ the function, and f the coordinate on which the group acts. Here, what we're looking for is $\hat{f}(\hat{x}) = g \cdot f(x)$. To find it, we have to introduce it somewhere in equation 160. For this reason we'll consider the group action acting the graph of f, $f(x)$:

$$(\hat{x}, \hat{f}) = \Big(X_g(x, f(x)), F_g(x, f(x)) \Big) \tag{161}$$

At this point, we have to introduce some notation to simplify things. Since both functions are really just functions of x, we can make that explicit:

$$(\hat{x}, \hat{f}) = \Big(X_g \circ (\mathrm{Id} \times f) \circ (x), F_g \circ (\mathrm{Id} \times f) \circ (x) \Big) \tag{162}$$

Here, $\mathrm{Id}(x) = x$, and $\times$ is the cartesian product. Written explicitly, we have $(\mathrm{Id} \times f) \circ (x) = (x, f(x))$. Using the second component, we have

$$\hat{f}(\hat{x}) = F_g \circ (\mathrm{Id} \times f) \circ (x) \tag{163}$$

We want this as a function of $\hat{x}$, so we also need to replace x using the first component of equation 161, when it is possible (otherwise this method doesn't work):

$$\hat{x} = X_g \circ (\mathrm{Id} \times f) \circ (x) \implies x = \Big[X_g \circ (\mathrm{Id} \times f) \Big]^{-1} \circ (\hat{x}) \tag{164}$$

Putting everything together gives

$$\hat{f}(\hat{x}) = F_g \circ (\mathrm{Id} \times f) \circ \Big[X_g \circ (\mathrm{Id} \times f) \Big]^{-1} \circ (\hat{x}) = F_g \circ X_g^{-1} \circ (\hat{x}) \tag{165}$$

This method will be enough to solve the Heat equation. The reader interested in learning more advanced methods is encouraged to look at the sources, all of which include a fuller treatment of the subject. Moreover, another DRP project by Louis Meunier (available on the DRP webpage) explores the link between symmetries of a special class of PDEs (variational problems) and conservation laws, through Noether's theorem. His text also examines various other topics related to partial differential equations and symmetries. The reader is encouraged to read that if interested.

Example – Solving the Heat Equation using symmetries

This example is adapted from example 2.41 of [Olv93], p.117. Consider the heat equation with independent dimensions x, t and dependent variable $u(x,t)$:

$$u_{xx} = u_t \tag{166}$$

We'll first find symmetries of this equation using the linearized symmetry condition. Doing so is an incredibly tedious but straightforward process, and the reader is encouraged to read the source material for all the details. The generators are the following:

$$\begin{aligned} X_1 &= \partial_x \\ X_2 &= \partial_t \\ X_3 &= u\partial_u \\ X_4 &= x\partial_x + 2t\partial_t \\ X_5 &= 2t\partial_x - xu\partial_u \\ X_6 &= 4tx\partial_x + 4t^2\partial_t - (x^2 + 2t)u\partial_u \end{aligned} \tag{167}$$

There's also an infinite dimensional algebra that encodes linearity:

$$X_\alpha = \alpha(x,t)\partial_u \tag{168}$$

where α is an arbitrary solution of the heat equation. The only two generators that we'll need are X_2 and X_6. Their group actions can be easily computed using any of the methods detailed previously (see section 2.5):

$$\begin{aligned} G_2 &: (x,t,u) \mapsto (x, t+\varepsilon, u) \\ G_6 &: (x,t,u) \mapsto \left(\frac{x}{1-4\varepsilon t}, \frac{t}{1-4\varepsilon t}, u\sqrt{1-4\varepsilon t}\, \exp\left(\frac{-\varepsilon x^2}{1-4\varepsilon t} \right) \right) \end{aligned} \tag{169}$$

Starting from the constant solution C, we'll find that the fundamental solution is a group-invariant solution of these two groups. We can apply G_6:

$$C \mapsto C\sqrt{1+4\varepsilon t}\, \exp\left(\frac{-\varepsilon x^2}{1-4\varepsilon t} \right) \tag{170}$$

As mentioned above, we want this function in the transformed coordinates $\hat{x}, \hat{t}$. We can invert the first two component of G_6 to get the following:

$$\frac{C}{\sqrt{1+4\varepsilon\hat{t}}} \exp\left(\frac{-\varepsilon\hat{x}^2}{1+4\varepsilon\hat{t}}\right) \tag{171}$$

We'll abuse notation a little and rename $\hat{x}, \hat{t}$ to x, t. To get the fundamental solution, we let $C = \sqrt{\varepsilon/\pi}$ and also rearrange the exponent:

$$\frac{1}{\sqrt{\pi(1/\varepsilon+4t)}} \exp\left(\frac{-x^2}{1/\varepsilon+4t}\right) \tag{172}$$

Then, the last step is to use G_2 and translate t by $-1/(4\varepsilon)$:

$$\frac{1}{\sqrt{4\pi t}} \exp\left(\frac{-x^2}{4t}\right) \tag{173}$$

And without that much effort, we have the fundamental solution of the heat equation!

A The Method of Characteristics

(This section is an adaptation of chapter 2 of [Cho22])

The method of characteristics is a technique to solve first order PDEs by reducing the equation to a system of ODEs along particular curves in the domain. Here we'll focus on linear PDEs, as they were the types of PDEs encountered in this text.

Consider a homogeneous equation of the form

$$a(x,y)u_x + b(x,y)u_y = 0 \tag{174}$$

that we're trying to solve on some domain $\Omega \subset \mathbb{R}^2$, given some initial data on a curve Γ in the domain. Notice that this is a statement about the directional derivative of $u(x,y)$. Indeed, the derivative of $u(x,y)$ along the vector $a(x,y), b(x,y)$ is simply 0:

$$\big(a(x,y), b(x,y)\big) \cdot \nabla u = 0 \tag{175}$$

At every point there exists a direction along which $u(x,y)$ stays constant. If we took a curve that followed that direction, or said differently, if we had a curve $(x(s), y(s))$ that had its tangent vector equal to $\big(a[x(s), y(s)], b[x(s), y(s)]\big)$, then $u(x(s), y(s))$ would be a constant along that curve. This curve is called a *characteristic curve* of the PDE.

The fact that $u(x,y)$ is constant along the characteristic curves means the following

$$\frac{d}{ds} u\big[x(s), y(s)\big] = 0 \tag{176}$$

$$\frac{dx}{ds} u_x + \frac{dy}{ds} u_y = 0 \tag{177}$$

From the PDE equation, we can deduce that along the characteristic curves the following holds:

$$\frac{dx}{ds} = a\big(x(s), y(s)\big) \quad \frac{dy}{ds} = b\big(x(s), y(s)\big) \tag{178}$$

These are the equations that can tell us what the characteristic curves are, and are aptly called the characteristic equations. Once we have those, we need to use the initial conditions given on Γ. Say that on it, $u(x,y) = g(x,y)$ for some arbitrary function g (smooth enough) of the curve Γ. Assume that every characteristic curve intersect Γ at one point, and call this point (x_0, y_0).

This gives an initial point for the characteristic curves: $(x(0), y(0)) = (x_0, y_0)$. Then, we also have that at $s = 0$, $u(x_0, y_0) = g(x_0, y_0)$. But $u(x,y)$ is constant along the characteristics! So we immediately have that $u(x(s), y(s)) = u(x,y) = g(x_0, y_0)$, for the entire characteristic curve. Using all characteristic curves together, we should be able to recover u on Ω. Indeed, we can always write $g(x_0, y_0)$ as a function of x, y, which gives the answer for the whole domain. Let's do an example to see how this works.

Example – A Simple Linear PDE

Consider the equation

$$u_x + yu_y = 0 \tag{179}$$

on the plane $y \geq 1$, with initial conditions given on the line $y = 1$: $u(x, 1) = x$. First, let's find the characteristic curves

$$\frac{dx}{ds} = 1 \quad \frac{dy}{ds} = y(s) \tag{180}$$

Solving this system is quite easy. We have $x(s) = s + x_0$ and $y(s) = y_0 e^s$. Using the initial conditions, we have that $y(s) = e^s$ (since $y(0) = 1$), and we also have that $u(x(s), y(s)) = x_0$. Let's isolate for x_0 then:

$$x_0 = x(s) - s = x(s) - \ln[y(s)] \tag{181}$$

And thus we have that

$$u(x, y) = x - \ln y \tag{182}$$

Let's check that this is indeed correct. First, when $y = 1$ we get $u(x, 1) = x$, so the initial conditions are satisfied. Next:

$$u_x = 1 \quad u_y = -\frac{1}{y} \tag{183}$$

And indeed, we see that the PDE is satisfied:

$$u_x + yu_y = 1 + \Big(\frac{-1}{y}\Big)y = 1 - 1 = 0 \tag{184}$$

In this text, sometimes we also had an equation of the form

$$a(x, y)u_x + b(x, y)u_y = c(x, y) \tag{185}$$

We can tweak slightly our previous method. Now we have that

$$\frac{d}{ds}u\big[x(s), y(s)\big] = c\big(x(s), y(s)\big) \tag{186}$$

Denote by $z(s) = u\big[x(s), y(s)\big]$ the value of u along the curve. There are now three characteristic equations:

$$\frac{dx}{ds} = a\big(x(s), y(s)\big) \quad \frac{dy}{ds} = b\big(x(s), y(s)\big) \quad \frac{dz}{ds} = c\big(x(s), y(s)\big) \tag{187}$$

The function u is not constant along the characteristics anymore, but follows $z(s)$. When solving for $z(s)$, the initial value will be given by $g(x_0, y_0)$ and the general solution $u(x, y)$ will be $z(s)$, after we have written the initial conditions and s as functions of x, y.

Example – A Simple Non-Homogeneous Linear PDE

Let's now consider the equation

$$u_x + yu_y = x \tag{188}$$

with initial conditions given by $u(x, 1) = x$. As explained above, we first solve the characteristic equations:

$$\frac{dx}{ds} = 1 \quad \frac{dy}{ds} = y(s) \quad \frac{dz}{ds} = x(s) \tag{189}$$

these are easily solved:

$$x(s) = s + x_0 \quad y(s) = y_0 e^s \quad z(s) = \frac{1}{2}s^2 + x_0 s + z_0 \tag{190}$$

The initial conditions enforce $z_0 = x_0$ and $y_0 = 1$. The equations become

$$x(s) = s + x_0 \quad y(s) = e^s \quad z(s) = \frac{1}{2}s^2 + x_0(1 + s) \tag{191}$$

We can then isolate x_0 and s:

$$x_0 = x - s \quad s = \ln y \tag{192}$$

Putting all of this together:

$$z = \frac{1}{2}\ln^2(y) + \big(x - \ln(y)\big)\big(1 + \ln(y)\big) \tag{193}$$

Which gives the solution.

This method can be easily generalized to n independent variables, by considering a domain in n dimensions and initial conditions given in an $n - 1$ dimensional subspace. For in-depth explanations, images and numerous examples, the reader is encouraged to see [Cho22].

References

[Cho22] Rustum Choksi. *Partial differential equations: a first course*, volume 54. American Mathematical Society, 2022.

[Hyd00] Peter Ellsworth Hydon. *Symmetry methods for differential equations: a beginner's guide*, volume 22. Cambridge University Press, 2000.

[Olv93] Peter J Olver. *Applications of Lie groups to differential equations*, volume 107. Springer Science & Business Media, 1993.